

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building a full-stack AI application (**Next.js** + Firebase/GCP) that surfaces guideline-backed medication alternatives in the EHR prescribing flow, taking features from design through implementation on a lean early-stage team.
- Extending a backend (Firebase Cloud Functions) that applies LLMs to clinical documents and guidelines, **RxNorm** normalization plus PubMed evidence, with **Gemini** constrained to structured synthesis while the backend deterministically controls results, keeping AI output grounded and verifiable.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building a full-stack platform (**Python**, **FastAPI**, sentence-transformers, **React**) that applies NLP to large unstructured text corpora, productionizing a published text-measurement method into a self-serve research tool.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**) with a reusable spreadsheet importer, and structured data-quality checks plus per-run reproducibility exports; grew the backend to **40 hermetic tests** in CI.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Architected a full-stack internal AI tool (React, Node.js, MongoDB) with secure role-based access, a training-data annotation platform used daily by the AI team that cut manual work by **100+ hours/month** and lifted labeling throughput **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly submissions** via REST APIs; contributed to **\$340K+** in revenue.
- Owned backend security and reliability for the multi-tenant stack, hardening authentication and rate limiting, adding **AWS KMS** encryption for sensitive data, and remediating **20+ VAPT findings** with regression-tested fixes.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage **agentic** web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research **agent** (TypeScript) that fetches web sources and streams multi-step research digests over WebSocket, with durable multi-step workflows that survive process restarts and tool failures and D1 for state.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built an **evaluation-improvement agent** that proposes queries and labels, scores them against multiple retrieval systems, and accepts only updates that improve discriminative power, with anti-Goodharting guardrails like read-only holdout protection.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash

LLM & Agents: LLM agents, RAG, MCP, tool use, prompt engineering, provider routing / fallback, evaluation, provenance

Full-Stack & Web: Next.js, React, TypeScript, FastAPI, Node.js, Express, Tailwind CSS, REST APIs

Data & Backend: PostgreSQL, MongoDB, Redis, SQLite, document parsing, structured extraction, auth / RBAC

Cloud & Deploy: AWS, GCP, Azure, Vercel, Docker, Git, GitHub Actions

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.